Exercises: Determinants, Orthogonal Matrices, and Orthonormal Bases – Answer Key Mathematical Foundations for Machine Learning

Determinant Properties and Volume Interpretation

Exercise 1.1 [Easy]

Let $A, B \in \mathbb{R}^{n\times n}$. Using only the determinant product rule $\det(AB) = \det(A)\det(B)$, prove the following properties:

(a) $\det(A^{-1}) = \frac{1}{\det(A)}$ for invertible A.
(b) $\det(\alpha A) = \alpha^n \det(A)$ for $\alpha \in \mathbb{R}$.
(c) If A is orthogonal, then $|\det(A)| = 1$.

Solution

(a) Since $AA^{-1} = I$, applying the product rule:

$$\det(A)\det(A^{-1}) = \det(I) = 1$$

Dividing both sides by $\det(A) \neq 0$ (since A is invertible) gives $\det(A^{-1}) = \frac{1}{\det(A)}$.
(b) We write $\alpha A = (\alpha I)A$, where αI is diagonal with all entries α. Then:

$$\det(\alpha A) = \det(\alpha I)\det(A) = \alpha^n \det(A)$$

since $\det(\alpha I) = \alpha^n$ (product of diagonal entries).
(c) Orthogonal means $A^T A = I$. Using the product rule and $\det(A^T) = \det(A)$:

$$\det(A^T)\det(A) = \det(I) = 1 \implies \det(A)^2 = 1 \implies |\det(A)| = 1.$$

Exercise 1.2 [Easy]

Let $R \in \mathbb{R}^{4\times 4}$ be an upper triangular matrix with diagonal entries $r_{11} = 2$, $r_{22} = -3$, $r_{33} = 1$, $r_{44} = 5$.

(a) What is $\det(R)$?
(b) What is $\det(R^T R)$?
(c) Is R invertible? If yes, what is $\det(R^{-1})$?

Solution

(a) For a triangular matrix, the determinant is the product of diagonal entries:

$$\det(R) = 2 \cdot (-3) \cdot 1 \cdot 5 = -30.$$

(b) Using $\det(AB) = \det(A)\det(B)$ and $\det(R^T) = \det(R)$:

$$\det(R^T R) = \det(R^T)\det(R) = (-30)(-30) = 900.$$

(c) Since $\det(R) = -30 \neq 0$, R is invertible. Then:

$$\det(R^{-1}) = \frac{1}{\det(R)} = -\frac{1}{30}.$$

Exercise 1.3 [Medium]

Consider the matrix

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}.$$

(a) Compute $\det(A)$. Interpret this geometrically: if $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ form the unit square, what is the area of the parallelogram spanned by Av_1 and Av_2?

(b) Let $B = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$. Without computing AB explicitly, what is $\det(AB)$? What is the volume scaling factor when applying first B then A?

(c) In a 2D generative model, a linear transformation T maps a latent distribution supported on the unit square to the output space. If T has $\det(T) = 0.5$, explain what this means for the density of the output distribution relative to a uniform input density. (Hint: The change-of-variables formula involves $|\det(J)|^{-1}$ for the density.)

Solution

(a) $\det(A) = 2 \cdot 3 - 1 \cdot 0 = 6$.

Geometrically, $Av_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ and $Av_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$. The area of the parallelogram spanned by these vectors equals $|\det(A)| = 6$. The unit square (area 1) is mapped to a parallelogram of area 6.

(b) By the product rule: $\det(AB) = \det(A)\det(B) = 6 \cdot (3 - 8) = 6 \cdot (-5) = -30$.

The volume scaling factor is $|\det(AB)| = 30$. The sign indicates orientation reversal.

(c) Under the change-of-variables formula for densities, if $y = T(x)$ and x has density $p_x(x)$, then:

$$p_y(y) = p_x(T^{-1}(y)) \cdot |\det(T)|^{-1}.$$

Since $|\det(T)| = 0.5$, the output density is scaled by $|\det(T)|^{-1} = 2$. The transformation compresses area by a factor of 0.5, so the same probability mass is concentrated in half the area, doubling the density. This is exactly the principle behind normalizing flows.

Exercise 1.4 [Medium]

Let $A \in \mathbb{R}^{n \times n}$ be invertible and let $u, v \in \mathbb{R}^n$. The **matrix determinant lemma** states:

$$\det(A + uv^T) = (1 + v^T A^{-1} u)\det(A).$$

(a) Using this lemma, compute $\det(I_n + uv^T)$ for arbitrary $u, v \in \mathbb{R}^n$.

(b) Let $v \in \mathbb{R}^n$ with $\|v\|_2 = 1$. Compute $\det(I_n + vv^T)$ and $\det(I_n - vv^T)$.

(c) In ridge regression, the regularized solution involves $(X^T X + \lambda I)^{-1}$. Suppose $X = uv^T$ is a rank-1 data matrix. Use part (a) to express $\det(X^T X + \lambda I)$ in terms of λ, u, and v.

Solution

(a) Setting $A = I_n$ (so $A^{-1} = I_n$):

$$\det(I_n + uv^T) = (1 + v^T I_n^{-1} u)\det(I_n) = 1 + v^T u = 1 + \langle v, u \rangle.$$

(b) For $\det(I_n + vv^T)$: set $u = v$ in part (a):

$$\det(I_n + vv^T) = 1 + v^T v = 1 + \|v\|_2^2 = 1 + 1 = 2.$$

For $\det(I_n - vv^T)$: set $u = -v$ in part (a):

$$\det(I_n - vv^T) = 1 + v^T(-v) = 1 - \|v\|_2^2 = 1 - 1 = 0.$$

This makes sense: $I_n - vv^T$ is the orthogonal projection onto $v^\perp$, which has rank $n-1$ and is therefore singular.

(c) We have $X^TX = (uv^T)^T(uv^T) = vu^Tuv^T = \|u\|^2vv^T$. So:

$$X^TX + \lambda I = \lambda I + \|u\|^2vv^T = \lambda\left(I + \frac{\|u\|^2}{\lambda}vv^T\right).$$

Using the scaling rule $\det(\alpha M) = \alpha^n \det(M)$ and part (a) with $\tilde{u} = \frac{\|u\|^2}{\lambda}v$:

$$\det(X^TX + \lambda I) = \lambda^n \det\left(I + \frac{\|u\|^2}{\lambda}vv^T\right) = \lambda^n\left(1 + \frac{\|u\|^2}{\lambda}\|v\|^2\right) = \lambda^n + \lambda^{n-1}\|u\|^2\|v\|^2.$$

Exercise 1.5 [Hard]

Let $A \in \mathbb{R}^{n\times n}$ be invertible. We define the map $T_A : \mathbb{R}^n \to \mathbb{R}^n$ by $T_A(x) = Ax$.

(a) Let $S \subset \mathbb{R}^n$ be a measurable set with volume $\mathrm{vol}(S)$. Show that $\mathrm{vol}(T_A(S)) = |\det(A)| \cdot \mathrm{vol}(S)$. You may use the substitution formula for integrals: $\int_{T_A(S)} 1\,dy = \int_S |\det(A)|\,dx$.

(b) Suppose $A = QR$ is the QR decomposition where Q is orthogonal and R is upper triangular with positive diagonal entries $r_{11}, \ldots, r_{nn}$. Show that $|\det(A)| = \prod_{i=1}^n r_{ii}$.

(c) In normalizing flows, one composes K invertible layers f_k with Jacobians J_k. Using the chain rule for determinants, express $\log|\det(J_{\text{total}})|$ in terms of the individual Jacobians. Why is the triangular Jacobian structure (as in autoregressive flows) computationally attractive?

Solution

(a) Using the substitution formula with the substitution $y = Ax$ (so $dy = |\det(A)|dx$):

$$\mathrm{vol}(T_A(S)) = \int_{T_A(S)} 1\,dy = \int_S |\det(A)|\,dx = |\det(A)|\int_S 1\,dx = |\det(A)| \cdot \mathrm{vol}(S).$$

(b) Since $A = QR$:

$$|\det(A)| = |\det(Q)\det(R)| = |\det(Q)| \cdot |\det(R)|.$$

Q is orthogonal so $|\det(Q)| = 1$. R is upper triangular so $\det(R) = \prod_{i=1}^n r_{ii}$. Since all $r_{ii} > 0$, we have $|\det(R)| = \prod_{i=1}^n r_{ii}$. Therefore:

$$|\det(A)| = \prod_{i=1}^n r_{ii}.$$

(c) The total transformation is $f = f_K \circ \cdots \circ f_1$. By the chain rule, $J_{\text{total}} = J_K \cdots J_1$, so:

$$\det(J_{\text{total}}) = \prod_{k=1}^K \det(J_k).$$

Taking logarithms:

$$\log|\det(J_{\text{total}})| = \sum_{k=1}^K \log|\det(J_k)|.$$

For autoregressive flows, each J_k is triangular, so $\det(J_k) = \prod_i (J_k)_{ii}$ and:

$$\log|\det(J_k)| = \sum_i \log|(J_k)_{ii}|,$$

which is $O(n)$ per layer instead of the $O(n^3)$ cost of computing a general determinant. This makes training with the change-of-variables loss tractable for high-dimensional data.

Orthonormal Bases and Orthogonal Matrices

Exercise 2.1 [Easy]

Let $Q \in \mathbb{R}^{n\times n}$ be an orthogonal matrix.

(a) Prove that Q preserves inner products: $\langle Qu, Qv\rangle = \langle u, v\rangle$ for all $u, v \in \mathbb{R}^n$.

(b) Prove that Q preserves norms: $\|Qu\|_2 = \|u\|_2$ for all $u \in \mathbb{R}^n$.

(c) If λ is an eigenvalue of Q (possibly complex), show that $|\lambda| = 1$.

Solution

(a) Using $Q^TQ = I$:

$$\langle Qu, Qv\rangle = (Qu)^T(Qv) = u^TQ^TQv = u^TIv = u^Tv = \langle u, v\rangle.$$

(b) Setting $v = u$ in part (a):

$$\|Qu\|_2^2 = \langle Qu, Qu\rangle = \langle u, u\rangle = \|u\|_2^2.$$

Taking square roots gives $\|Qu\|_2 = \|u\|_2$.

(c) Let $Qv = \lambda v$ for some $v \neq 0$ (where $\lambda \in \mathbb{C}$ and $v \in \mathbb{C}^n$ in general). Taking norms:

$$\|Qv\|_2 = |\lambda| \cdot \|v\|_2.$$

But Q preserves norms (this extends to the complex inner product for unitary matrices, and Q viewed over $\mathbb{C}$ is unitary), so $\|Qv\|_2 = \|v\|_2$. Since $v \neq 0$, dividing by $\|v\|_2 > 0$ gives $|\lambda| = 1$.

Exercise 2.2 [Medium]

Consider $\mathbb{R}^3$ with the standard inner product. Let

$$v_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\\0\end{pmatrix}, \quad v_2 = \frac{1}{\sqrt{6}}\begin{pmatrix}1\\-1\\2\end{pmatrix}, \quad v_3 = \frac{1}{\sqrt{3}}\begin{pmatrix}-1\\1\\1\end{pmatrix}.$$

(a) Verify that $\{v_1, v_2, v_3\}$ is an orthonormal basis of $\mathbb{R}^3$.

(b) Express $w = \begin{pmatrix}3\\1\\2\end{pmatrix}$ in the basis $\{v_1, v_2, v_3\}$, i.e., find coefficients $\alpha_1, \alpha_2, \alpha_3$ such that $w = \alpha_1v_1 + \alpha_2v_2 + \alpha_3v_3$.

(c) Let $Q = (v_1 \mid v_2 \mid v_3)$ be the matrix with columns v_1, v_2, v_3. Verify that Q^Tw gives the coordinate vector of w in this basis. Why does this work?

Solution

(a) We check orthonormality. Norms:

$$\|v_1\|^2 = \frac{1}{2}(1+1+0) = 1, \quad \|v_2\|^2 = \frac{1}{6}(1+1+4) = 1, \quad \|v_3\|^2 = \frac{1}{3}(1+1+1) = 1.$$

Inner products:

$$\langle v_1, v_2\rangle = \frac{1}{\sqrt{12}}(1-1+0) = 0, \quad \langle v_1, v_3\rangle = \frac{1}{\sqrt{6}}(-1+1+0) = 0,$$

$$\langle v_2, v_3\rangle = \frac{1}{\sqrt{18}}(-1-1+2) = 0.$$

All pairwise orthogonal and unit norm, so this is an ONB of $\mathbb{R}^3$.
(b) For an ONB, $\alpha_i = \langle w, v_i \rangle$:

$$\alpha_1 = \langle w, v_1 \rangle = \frac{1}{\sqrt{2}}(3+1+0) = \frac{4}{\sqrt{2}} = 2\sqrt{2},$$

$$\alpha_2 = \langle w, v_2 \rangle = \frac{1}{\sqrt{6}}(3-1+4) = \frac{6}{\sqrt{6}} = \sqrt{6},$$

$$\alpha_3 = \langle w, v_3 \rangle = \frac{1}{\sqrt{3}}(-3+1+2) = 0.$$

So $w = 2\sqrt{2}\, v_1 + \sqrt{6}\, v_2 + 0 \cdot v_3$.
(c) Computing $Q^T w$: the i-th entry of $Q^T w$ is $v_i^T w = \langle v_i, w \rangle = \alpha_i$. So:

$$Q^T w = \begin{pmatrix} 2\sqrt{2} \\ \sqrt{6} \\ 0 \end{pmatrix}.$$

This works because Q is orthogonal ($Q^{-1} = Q^T$), so if $w = Q\alpha$ (where α is the coordinate vector), then $\alpha = Q^{-1}w = Q^T w$.

Exercise 2.3 [Medium]

Let $Q \in \mathbb{R}^{n \times n}$ be orthogonal and $A \in \mathbb{R}^{n \times n}$ be symmetric.

(a) Show that $Q^T A Q$ is symmetric.

(b) Show that A and $Q^T A Q$ have the same eigenvalues.

(c) Show that $\|Ax\|_2 = \|Q^T A Q y\|_2$ when $x = Qy$. Interpret: does the orthogonal change of basis affect the "behavior" of the linear map A?

Solution

(a) We compute the transpose of $Q^T A Q$:

$$(Q^T A Q)^T = Q^T A^T (Q^T)^T = Q^T A^T Q.$$

Since A is symmetric ($A^T = A$), this equals $Q^T A Q$. So $Q^T A Q$ is symmetric.
(b) Let λ be an eigenvalue of A with eigenvector v: $Av = \lambda v$. Set $w = Q^T v$ (equivalently $v = Qw$). Then:

$$(Q^T A Q)w = Q^T A(Qw) = Q^T A v = Q^T(\lambda v) = \lambda Q^T v = \lambda w.$$

So λ is also an eigenvalue of $Q^T A Q$ with eigenvector $w = Q^T v$. Since Q is invertible, this is a bijection between eigenpairs, so the eigenvalues are identical.
(c) With $x = Qy$:

$$\|Ax\|_2 = \|AQy\|_2 = \|QQ^T A Q y\|_2 = \|Q^T A Q y\|_2,$$

where the last equality uses the fact that Q preserves norms. The orthogonal change of basis does not change the "behavior" of A: norms of outputs, eigenvalues, determinant, trace, and singular values are all preserved. We are simply expressing the same linear map in a different orthonormal coordinate system.

Exercise 2.4 [Hard]

Let $P \in \mathbb{R}^{n\times n}$ be a matrix satisfying $P^2 = P$ and $P^T = P$ (i.e., P is an orthogonal projection).

(a) Show that the only possible eigenvalues of P are 0 and 1.

(b) Let $r = \text{rank}(P)$. Show that $\text{tr}(P) = r$.

(c) Show that $I - P$ is also an orthogonal projection. What subspace does it project onto?

(d) In the context of PCA, suppose we project data onto the top-r principal components. The projection matrix is $P = U_r U_r^T$ where $U_r \in \mathbb{R}^{n\times r}$ has orthonormal columns. Verify that $P^2 = P$ and $P^T = P$, and show that $\text{tr}(P) = r$.

Solution

(a) Let $Pv = \lambda v$ with $v \neq 0$. Applying P again:

$$P^2 v = P(\lambda v) = \lambda P v = \lambda^2 v.$$

But $P^2 = P$, so $Pv = \lambda v$ as well. Therefore $\lambda^2 v = \lambda v$, giving $(\lambda^2 - \lambda)v = 0$. Since $v \neq 0$: $\lambda^2 - \lambda = 0 \implies \lambda(\lambda - 1) = 0$. So $\lambda \in \{0, 1\}$.

(b) Since P is symmetric, it is diagonalizable with real eigenvalues. By part (a), all eigenvalues are either 0 or 1. The rank of P equals the number of nonzero eigenvalues (counted with multiplicity), which is the number of eigenvalues equal to 1. Call this r. Then:

$$\text{tr}(P) = \sum_{i=1}^{n} \lambda_i = r \cdot 1 + (n - r) \cdot 0 = r.$$

(c) We verify both conditions:

- Idempotent: $(I - P)^2 = I - 2P + P^2 = I - 2P + P = I - P$. ✓
- Symmetric: $(I - P)^T = I^T - P^T = I - P$. ✓

So $I - P$ is an orthogonal projection. It projects onto $\ker(P) = (\text{ran}(P))^\perp$, the orthogonal complement of the range of P. Indeed, for any x: $(I - P)x = x - Px \in \ker(P)$ since $P(x - Px) = Px - P^2 x = 0$.

(d) Symmetric: $(U_r U_r^T)^T = (U_r^T)^T U_r^T = U_r U_r^T$. ✓

Idempotent: $(U_r U_r^T)(U_r U_r^T) = U_r (U_r^T U_r) U_r^T = U_r I_r U_r^T = U_r U_r^T$, since U_r has orthonormal columns so $U_r^T U_r = I_r$. ✓

Trace: $\text{tr}(U_r U_r^T) = \text{tr}(U_r^T U_r) = \text{tr}(I_r) = r$, using the cyclic property $\text{tr}(AB) = \text{tr}(BA)$.

Gram-Schmidt, Orthogonal Projection, QR Decomposition, and Change of Basis

Exercise 3.1 [Medium]

Apply the Gram-Schmidt process to the vectors

$$u_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad u_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad u_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

to obtain an orthonormal basis $\{v_1, v_2, v_3\}$ of $\mathbb{R}^3$.

(a) Compute v_1.

(b) Compute v_2 by removing the component of u_2 along v_1 and normalizing.

(c) Compute v_3 by removing the components of u_3 along v_1 and v_2, then normalizing.

(d) Verify that the matrix $Q = (v_1 \mid v_2 \mid v_3)$ is orthogonal.

Solution

(a) $v_1 = \frac{u_1}{\|u_1\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

(b) Remove the projection of u_2 onto v_1:

$$\langle u_2, v_1 \rangle = \frac{1}{\sqrt{2}}(1 + 0 + 0) = \frac{1}{\sqrt{2}}.$$

$$v_2' = u_2 - \langle u_2, v_1 \rangle v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1/2 \\ 1/2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix}.$$

$$\|v_2'\| = \sqrt{1/4 + 1/4 + 1} = \sqrt{3/2}, \quad v_2 = \frac{v_2'}{\|v_2'\|} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

(c) Remove projections onto v_1 and v_2:

$$\langle u_3, v_1 \rangle = \frac{1}{\sqrt{2}}(0 + 1 + 0) = \frac{1}{\sqrt{2}}, \quad \langle u_3, v_2 \rangle = \frac{1}{\sqrt{6}}(0 - 1 + 2) = \frac{1}{\sqrt{6}}.$$

$$v_3' = u_3 - \frac{1}{\sqrt{2}} v_1 - \frac{1}{\sqrt{6}} v_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} - \frac{1}{6} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -2/3 \\ 2/3 \\ 2/3 \end{pmatrix}.$$

$$\|v_3'\| = \frac{2}{3}\sqrt{3}, \quad v_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}.$$

(d) We verify $Q^T Q = I$: the (i, j) entry of $Q^T Q$ is $\langle v_i, v_j \rangle = \delta_{ij}$ by construction. Since Q is square and $Q^T Q = I$, Q is orthogonal.

Exercise 3.2 [Medium]

Let $W = \operatorname{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\} \subset \mathbb{R}^3$.

(a) Find the orthogonal projection matrix P_W onto W. (Hint: If U has columns forming a basis of W, then $P_W = U(U^T U)^{-1} U^T$.)

(b) Compute the orthogonal projection of $b = \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix}$ onto W.

(c) Find the distance from b to the subspace W, i.e., $\|b - P_W b\|_2$.

(d) In least-squares regression, $\hat{y} = X(X^T X)^{-1} X^T y$ is the projection of y onto $\operatorname{col}(X)$. The residual $y - \hat{y}$ is orthogonal to $\operatorname{col}(X)$. Verify this orthogonality property for your computation above: check that $b - P_W b \perp w$ for each basis vector w of W.

Solution

(a) Let $U = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$. Then:

$$U^T U = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad (U^T U)^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

$$P_W = U(U^T U)^{-1} U^T = \frac{1}{3}\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

(b)

$$P_W b = \frac{1}{3}\begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}\begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 3 \\ 3 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

(c) The residual is:

$$b - P_W b = \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -2 \end{pmatrix}.$$

Distance: $\|b - P_W b\|_2 = \sqrt{4+4+4} = 2\sqrt{3}$.

(d) Let $w_1 = (1,0,1)^T$ and $w_2 = (0,1,1)^T$. The residual is $r = (2,2,-2)^T$.

$$\langle r, w_1 \rangle = 2 + 0 - 2 = 0 \quad \checkmark$$

$$\langle r, w_2 \rangle = 0 + 2 - 2 = 0 \quad \checkmark$$

The residual is orthogonal to both basis vectors of W, hence orthogonal to all of W, confirming the fundamental property of orthogonal projection (and of least-squares).

Exercise 3.3 [Medium]

Find the QR decomposition of the matrix

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

(a) Apply Gram-Schmidt to the columns of A to find the orthogonal matrix Q.
(b) Compute $R = Q^T A$ and verify it is upper triangular with positive diagonal.
(c) Verify that $|\det(A)| = \prod_i r_{ii}$.
(d) Use the QR decomposition to solve $Ax = b$ for $b = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ via back-substitution on $Rx = Q^T b$.

Solution

(a) The columns of A are $a_1 = (1,1,0)^T$, $a_2 = (1,0,1)^T$, $a_3 = (0,1,1)^T$. These are exactly

the vectors from Exercise 3.1. By Gram-Schmidt:

$$q_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\\0\end{pmatrix}, \quad q_2 = \frac{1}{\sqrt{6}}\begin{pmatrix}1\\-1\\2\end{pmatrix}, \quad q_3 = \frac{1}{\sqrt{3}}\begin{pmatrix}-1\\1\\1\end{pmatrix}.$$

(b) $R = Q^T A$. We compute using $r_{ij} = \langle q_i, a_j\rangle$:

$$R = \begin{pmatrix}\langle q_1,a_1\rangle & \langle q_1,a_2\rangle & \langle q_1,a_3\rangle\\ 0 & \langle q_2,a_2\rangle & \langle q_2,a_3\rangle\\ 0 & 0 & \langle q_3,a_3\rangle\end{pmatrix} = \begin{pmatrix}\sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ 0 & \sqrt{\frac{3}{2}} & \frac{1}{\sqrt{6}}\\ 0 & 0 & \frac{2}{\sqrt{3}}\end{pmatrix}.$$

All diagonal entries are positive. ✓
(c) $\prod_i r_{ii} = \sqrt{2}\cdot\sqrt{3/2}\cdot 2/\sqrt{3} = \sqrt{2}\cdot\sqrt{3/2}\cdot 2/\sqrt{3}$.
Simplify: $\sqrt{2}\cdot\frac{\sqrt{3}}{\sqrt{2}}\cdot\frac{2}{\sqrt{3}} = 2$.
Direct computation: $\det(A) = 1(0-1) - 1(1-0) + 0 = -2$. So $|\det(A)| = 2 = \prod_i r_{ii}$. ✓
(d) Compute $Q^T b$:

$$Q^T b = \begin{pmatrix}q_1^T b\\ q_2^T b\\ q_3^T b\end{pmatrix} = \begin{pmatrix}\frac{3}{\sqrt{2}}\\ \frac{7}{\sqrt{6}}\\ \frac{2}{\sqrt{3}}\end{pmatrix}.$$

Solve $Rx = Q^T b$ by back-substitution:
From row 3: $\frac{2}{\sqrt{3}}x_3 = \frac{2}{\sqrt{3}} \implies x_3 = 1$.
From row 2: $\sqrt{\frac{3}{2}}x_2 + \frac{1}{\sqrt{6}}\cdot 1 = \frac{7}{\sqrt{6}} \implies \sqrt{\frac{3}{2}}x_2 = \frac{6}{\sqrt{6}} = \sqrt{6} \implies x_2 = \sqrt{6}/\sqrt{3/2} = 2$.
From row 1: $\sqrt{2}x_1 + \frac{1}{\sqrt{2}}\cdot 2 + \frac{1}{\sqrt{2}}\cdot 1 = \frac{3}{\sqrt{2}} \implies \sqrt{2}x_1 = \frac{3}{\sqrt{2}} - \frac{3}{\sqrt{2}} = 0 \implies x_1 = 0$.
Solution: $x = (0,2,1)^T$.

Exercise 3.4 [Medium]

Let $B_1 = \{e_1, e_2\}$ be the standard basis of $\mathbb{R}^2$ and let $B_2 = \left\{w_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix},\ w_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}-1\\1\end{pmatrix}\right\}$.

(a) Verify that B_2 is an orthonormal basis.

(b) Find the transition matrix Q from B_1 to B_2 such that $[x]_{B_2} = Q[x]_{B_1}$. (Hint: $Q = W^T V$ where V, W have the respective basis vectors as columns.)

(c) Verify that Q is orthogonal and compute $\det(Q)$. What geometric transformation does Q represent?

(d) Given the vector $x = \begin{pmatrix}3\\1\end{pmatrix}$ in B_1, find its coordinates in B_2.

Solution

(a) $\|w_1\|^2 = \frac{1}{2}(1+1) = 1$, $\|w_2\|^2 = \frac{1}{2}(1+1) = 1$, $\langle w_1, w_2\rangle = \frac{1}{2}(-1+1) = 0$. So B_2 is an ONB. ✓
(b) Here $V = I_2$ (standard basis) and $W = (w_1 \mid w_2) = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & -1\\ 1 & 1\end{pmatrix}$. The transition matrix is:

$$Q = W^T V = W^T = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ -1 & 1\end{pmatrix}.$$

(c) Check: $Q^TQ = \frac{1}{2}\begin{pmatrix}1 & -1\\ 1 & 1\end{pmatrix}\begin{pmatrix}1 & 1\\ -1 & 1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix} = I$. ✓
$\det(Q) = \frac{1}{2}(1\cdot 1 - 1\cdot(-1)) = \frac{1}{2}\cdot 2 = 1$.
Since $\det(Q) = 1$ (in $SO(2)$), Q represents a rotation. Specifically, Q is a rotation by $-45°$ (or equivalently, B_2 is the standard basis rotated by $+45°$).
(d)

$$[x]_{B_2} = Q[x]_{B_1} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ -1 & 1\end{pmatrix}\begin{pmatrix}3\\ 1\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}4\\ -2\end{pmatrix} = \begin{pmatrix}2\sqrt{2}\\ -\sqrt{2}\end{pmatrix}.$$

So $x = 2\sqrt{2}\,w_1 + (-\sqrt{2})\,w_2$.

Mixed Challenging Exercises

Exercise 4.1 [Hard]

Let $A \in \mathbb{R}^{n\times n}$ be symmetric with eigendecomposition $A = Q\Lambda Q^T$, where Q is orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.

(a) Show that $\det(A) = \prod_{i=1}^n \lambda_i$ and $\operatorname{tr}(A) = \sum_{i=1}^n \lambda_i$.

(b) For the matrix $A = 2vv^T + 3I$ where $v \in \mathbb{R}^n$ with $\|v\| = 1$, find all eigenvalues, $\det(A)$, $\operatorname{tr}(A)$, and all singular values.

(c) Compute $\max_{\|x\|=1} x^TAx$ and $\max_{\|x\|=1} \|Ax\|_2$ for the matrix in part (b).

(d) Suppose we apply gradient descent on the quadratic $f(x) = \frac{1}{2}x^TAx$ with A from part (b). The condition number is $\kappa = \lambda_{\max}/\lambda_{\min}$. Compute κ and discuss how the convergence rate depends on it.

Solution

(a) Determinant: $\det(A) = \det(Q\Lambda Q^T) = \det(Q)\det(\Lambda)\det(Q^T) = \det(\Lambda) = \prod_i \lambda_i$, using $\det(Q)\det(Q^T) = \det(QQ^T) = \det(I) = 1$.
Trace: $\operatorname{tr}(A) = \operatorname{tr}(Q\Lambda Q^T) = \operatorname{tr}(Q^TQ\Lambda) = \operatorname{tr}(\Lambda) = \sum_i \lambda_i$, using the cyclic property.
(b) $A = 2vv^T + 3I$. For $x = v$: $Av = 2\|v\|^2 v + 3v = 5v$, so $\lambda = 5$ with eigenvector v.
For $x \perp v$: $Ax = 2vv^Tx + 3x = 0 + 3x = 3x$, so $\lambda = 3$ with multiplicity $n-1$.
Eigenvalues: 5 (once) and 3 ($n-1$ times).
$\det(A) = 5\cdot 3^{n-1}$.
$\operatorname{tr}(A) = 5 + 3(n-1) = 3n+2$.
Since A is symmetric PSD (all eigenvalues > 0), singular values equal eigenvalues: $\sigma_1 = 5$, $\sigma_2 = \cdots = \sigma_n = 3$.
(c) By the Rayleigh quotient:

$$\max_{\|x\|=1} x^TAx = \lambda_{\max} = 5, \quad \text{attained at } x = v.$$

$$\max_{\|x\|=1} \|Ax\|_2 = \sigma_{\max} = 5, \quad \text{attained at } x = v.$$

(d) $\kappa = \lambda_{\max}/\lambda_{\min} = 5/3$.
For gradient descent on a quadratic with step size $\eta = 1/\lambda_{\max}$, the error contracts at rate:

$$\|x_{k+1} - x^*\| \le \left(\frac{\kappa-1}{\kappa+1}\right)\|x_k - x^*\|.$$

Here the contraction factor is $\frac{5/3-1}{5/3+1} = \frac{2/3}{8/3} = \frac{1}{4}$. This is close to 0, indicating fast convergence.
In general, κ close to 1 (well-conditioned) gives fast convergence, while $\kappa \gg 1$ (ill-conditioned) gives slow convergence.

Exercise 4.2 [Hard]

Let $u_1, \ldots, u_n$ be an orthonormal basis of $\mathbb{R}^n$ and define $P_k = u_k u_k^T$ for each k.

(a) Show that $\sum_{k=1}^n P_k = I_n$ (the completeness relation / resolution of the identity).

(b) Show that $P_j P_k = 0$ for $j \neq k$ and $P_k^2 = P_k$.

(c) For a symmetric matrix $A = \sum_{k=1}^n \lambda_k P_k$ (spectral decomposition), derive $\det(A)$ and $\operatorname{tr}(A)$ using the properties from (a) and (b).

(d) Using the resolution of the identity, show that for any $x \in \mathbb{R}^n$:

$$\|x\|^2 = \sum_{k=1}^n |\langle x, u_k \rangle|^2$$

(Parseval's identity). Explain why this is relevant for signal compression in ML.

Solution

(a) For any $x \in \mathbb{R}^n$, expand x in the ONB: $x = \sum_{k=1}^n \langle x, u_k \rangle u_k = \sum_{k=1}^n (u_k u_k^T) x = \left(\sum_{k=1}^n P_k\right) x$.
Since this holds for all x, we conclude $\sum_{k=1}^n P_k = I_n$.
(b) $P_j P_k = (u_j u_j^T)(u_k u_k^T) = u_j (u_j^T u_k) u_k^T = u_j \langle u_j, u_k \rangle u_k^T$.
If $j \neq k$: $\langle u_j, u_k \rangle = 0$, so $P_j P_k = 0$.
If $j = k$: $\langle u_k, u_k \rangle = 1$, so $P_k^2 = u_k \cdot 1 \cdot u_k^T = P_k$.
(c) Determinant: We use the fact that $A = \sum_k \lambda_k P_k$ has eigenvalues λ_k (since $A u_k = \lambda_k u_k$).
Therefore $\det(A) = \prod_{k=1}^n \lambda_k$.
Trace: $\operatorname{tr}(A) = \operatorname{tr}(\sum_k \lambda_k P_k) = \sum_k \lambda_k \operatorname{tr}(P_k)$. Since $\operatorname{tr}(P_k) = \operatorname{tr}(u_k u_k^T) = u_k^T u_k = 1$, we get $\operatorname{tr}(A) = \sum_{k=1}^n \lambda_k$.
(d) Using $\sum_k P_k = I$:

$$\|x\|^2 = x^T x = x^T I x = x^T \left(\sum_{k=1}^n u_k u_k^T \right) x = \sum_{k=1}^n x^T u_k u_k^T x = \sum_{k=1}^n (u_k^T x)^2 = \sum_{k=1}^n |\langle x, u_k \rangle|^2.$$

Relevance: Parseval's identity says that the total energy (squared norm) of a signal equals the sum of squared coefficients in any ONB. For compression, we choose an ONB (e.g., PCA directions, wavelet basis) where most energy concentrates in a few coefficients $\langle x, u_k \rangle$. We keep only the $r \ll n$ largest coefficients and discard the rest; the reconstruction error is $\sum_{k=r+1}^n |\langle x, u_k \rangle|^2$, which is small if the basis is well-chosen.

Exercise 4.3 [Hard]

Let $X \in \mathbb{R}^{n \times d}$ be a data matrix with n samples and d features. The thin QR decomposition is $X = QR$ where $Q \in \mathbb{R}^{n \times d}$ has orthonormal columns and $R \in \mathbb{R}^{d \times d}$ is upper triangular.

(a) Show that the least-squares solution $\hat{\theta} = (X^T X)^{-1} X^T y$ can be rewritten as $\hat{\theta} = R^{-1} Q^T y$.

(b) Show that the projection matrix onto $\operatorname{col}(X)$ satisfies $P = QQ^T$ (i.e., it depends only on Q, not on R).

(c) Show that $\det(X^T X) = (\det R)^2 = \left(\prod_{i=1}^d r_{ii}\right)^2$.

(d) In numerical linear algebra, solving normal equations via $(X^T X)^{-1} X^T y$ can be numerically unstable because $\kappa(X^T X) = \kappa(X)^2$. Explain why the QR-based approach $\hat{\theta} = R^{-1} Q^T y$ is more stable.

Solution

(a) Substitute $X = QR$:

$$X^T X = (QR)^T (QR) = R^T Q^T Q R = R^T R,$$

since $Q^T Q = I_d$. Also $X^T y = R^T Q^T y$. So:

$$\hat{\theta} = (R^T R)^{-1} R^T Q^T y = R^{-1} (R^T)^{-1} R^T Q^T y = R^{-1} Q^T y.$$

(b) The projection is $P = X(X^T X)^{-1} X^T$. Substituting:

$$P = QR(R^T R)^{-1}(QR)^T = QR \cdot R^{-1}(R^T)^{-1} \cdot R^T Q^T = Q I_d Q^T = Q Q^T.$$

This is intuitive: the projection onto $\mathrm{col}(X) = \mathrm{col}(Q)$ only depends on the orthonormal basis Q for that column space, not on R (which encodes how the original columns are expressed in that basis).

(c) $\det(X^T X) = \det(R^T R) = \det(R^T)\det(R) = (\det R)^2$. Since R is upper triangular: $\det(R) = \prod_{i=1}^d r_{ii}$, so $\det(X^T X) = \left(\prod_{i=1}^d r_{ii}\right)^2$.

(d) The condition number satisfies $\kappa(X^T X) = \kappa(R^T R) = \kappa(R)^2$. Forming $X^T X$ squares the condition number, amplifying rounding errors. The QR approach instead:

1. Computes $Q^T y$ (orthogonal multiplication, stable: $\kappa(Q) = 1$).
2. Solves $R\hat{\theta} = Q^T y$ by back-substitution (triangular solve with condition number $\kappa(R) = \kappa(X)$, not $\kappa(X)^2$).

By avoiding the squaring of the condition number, the QR approach retains roughly twice as many digits of accuracy.